AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 26 · STUDENT LESSON

Evaluation Sets

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Freeze the cases. Score the behavior. Inspect the gaps.

Evaluate answerable and missing-evidence behavior without hiding failures in a pooled rate.

Keep one habit: Case contract → six-case mixture change → complete ten-case practice → coverage audit → independent club gaps.

MAIN LESSON · SECTION 1

Which cases would expose a failure that an easy example hides?

> Separate case coverage, group outcomes and pooled totals.

Your learning goals

I can explain the **mechanism** and terminology of system **evaluation set**s.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Recall criteria and ranking limits

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall Quality Rubrics: declare observable criteria and mandatory gates before inspecting outputs. Record each case's **evidence** rather than treating a total as a complete quality judgement.

- Recall Week 18's retrieval extension: a ranking score is not relevance or source truth. This week's optional extension gives explicit judged sets and cutoffs for precision and recall.

Your end goal

By the end, I can explain **evaluation set**s using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Quality Rubrics. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Block to Python Bridge. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Conversation State and Memory. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 26: **Evaluation Set**s

A good eval set represents real use, edge cases, and known risks.



CREATOR MISSION

Author ten tests with **expected behavior**.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Readiness for Ranking and retrieval evaluation

> **My readiness explanation**

This week’s end goal

MAIN LESSON · SECTION 2

Choose cases from the task and its failure risks

> Expected behavior must be set before viewing system outputs.

Understand coverage and case roles

An evaluation set is a recorded collection of inputs, permitted evidence and expected behavior. Include answerable, missing-source, ambiguous, conflicting and access-boundary cases that matter for the task. Development cases guide repairs; final reserved cases are kept from tuning. This lesson’s visible examples are practice, so they cannot serve as a hidden final test.

A small constructed set probes specific behavior; it is not a representative estimate of all future requests.

A system **evaluation set** contains cases with expected behaviour and scoring rules. It should cover retrieval, support, fallback and permissions rather than only easy answerable questions.

▼ **Extension · Ranking needs relevance judgements**

A ranking needs task-specific relevance judgments and a declared **cutoff**. Precision and recall have different denominators.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use **evaluation set**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Apply the connected idea: Ranking and retrieval evaluation

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Freeze expected outcomes before measuring performance

> Different denominators answer different questions.

Complete case-record contract

Case record	What to freeze before scoring
Identity	Case ID, user question and requested time
Evidence	Eligible source IDs/versions and relevant passages
Expected behavior	Supported claim, clarification, fallback or access refusal
Rubric	Required conditions and mandatory failures
Exposure	Development/reviewed or reserved; who has seen it
Observed result	Actual supplied output or recorded run, distinct from prediction
Aggregation	Report subgroup numerator/denominator before the pooled rate

Record case ID, **evidence** version, expected claims or fallback, permitted actions and scoring rubric. Separate development from final cases. Aggregate metrics only after checking required conditions individually.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the deciding check

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Extension · Precision and recall definitions

Precision at k is relevant returned items divided by k when exactly k items are returned.
Recall at k is relevant returned items divided by all known relevant eligible items.

If the relevant set is unknown, do not invent recall.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

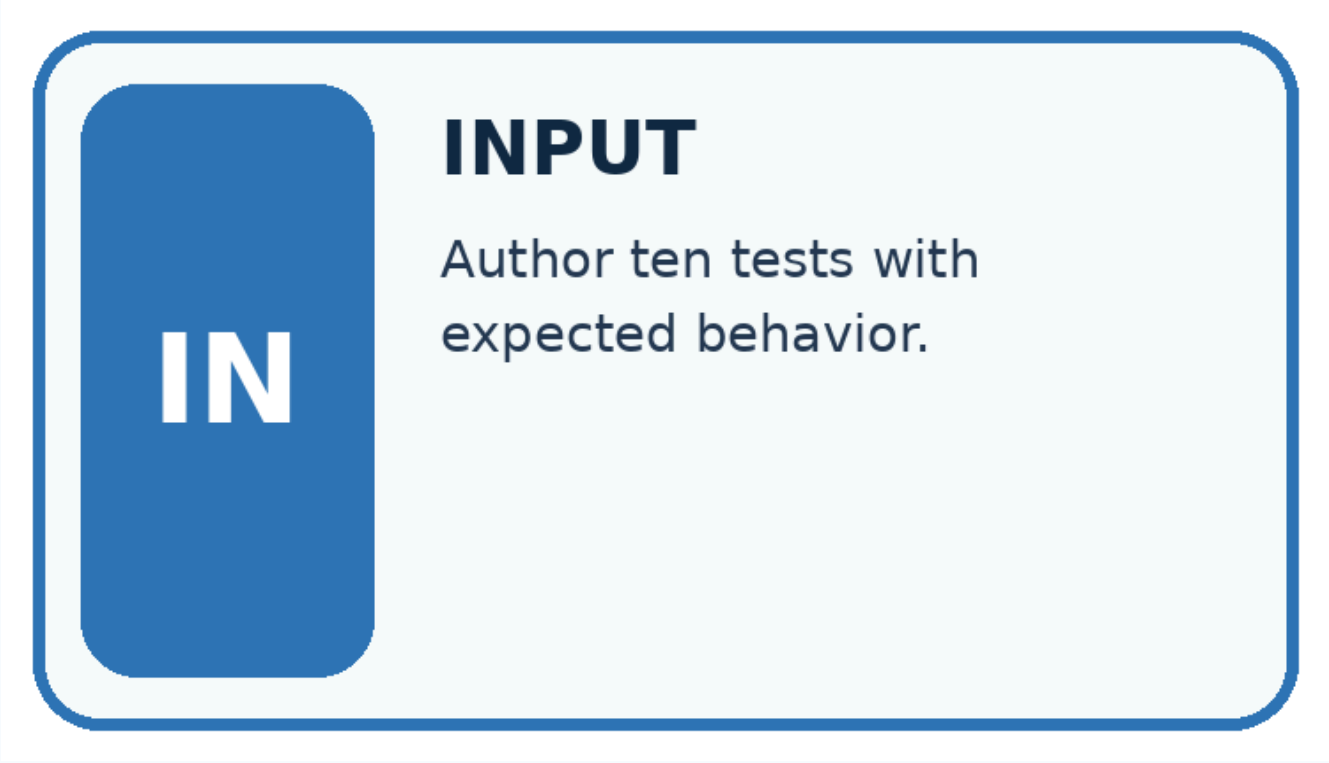
If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

▼ **Earlier wording, alternatives and picture transcripts**

VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.



↓


SYSTEM
A good eval set represents real use, edge cases, and known risks.
↓
OUT
EVIDENCE
Author ten tests with expected behavior.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

System evaluation sets — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Ten cases include six answerable and four missing-evidence cases. System answers five answerable correctly and abstains appropriately on three missing cases.	Record case ID, evidence version, expected claims or fallback, permitted actions and scoring rubric. Separate development from final cases. Aggregate metrics only after checking required conditions individually.	Answer correctness 5/6; appropriate fallback 3/4; combined contract passes 8/10 if each case has one required outcome. Different denominators answer different questions.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Apply the connected idea: Ranking and retrieval evaluation

Ranking and retrieval evaluation — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Relevant documents are A, C, E. Top three results A, B, C contain two relevant items.	Precision at k is relevant returned items divided by k when exactly k items are returned.	Precision@3=2/3; recall@3=2/3. Duplicates should not count as distinct retrieved documents.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

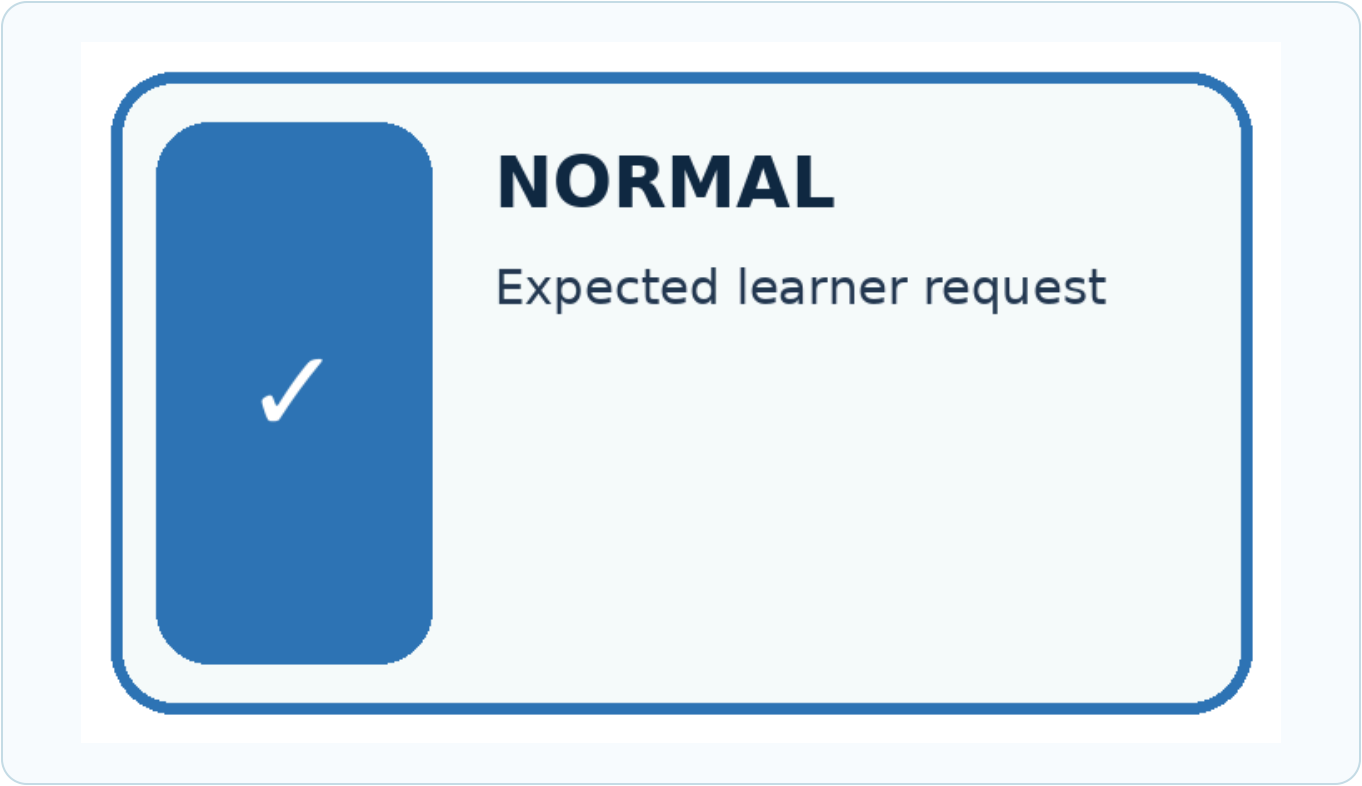
Connect the picture and the explanation

MAIN LESSON · SECTION 4

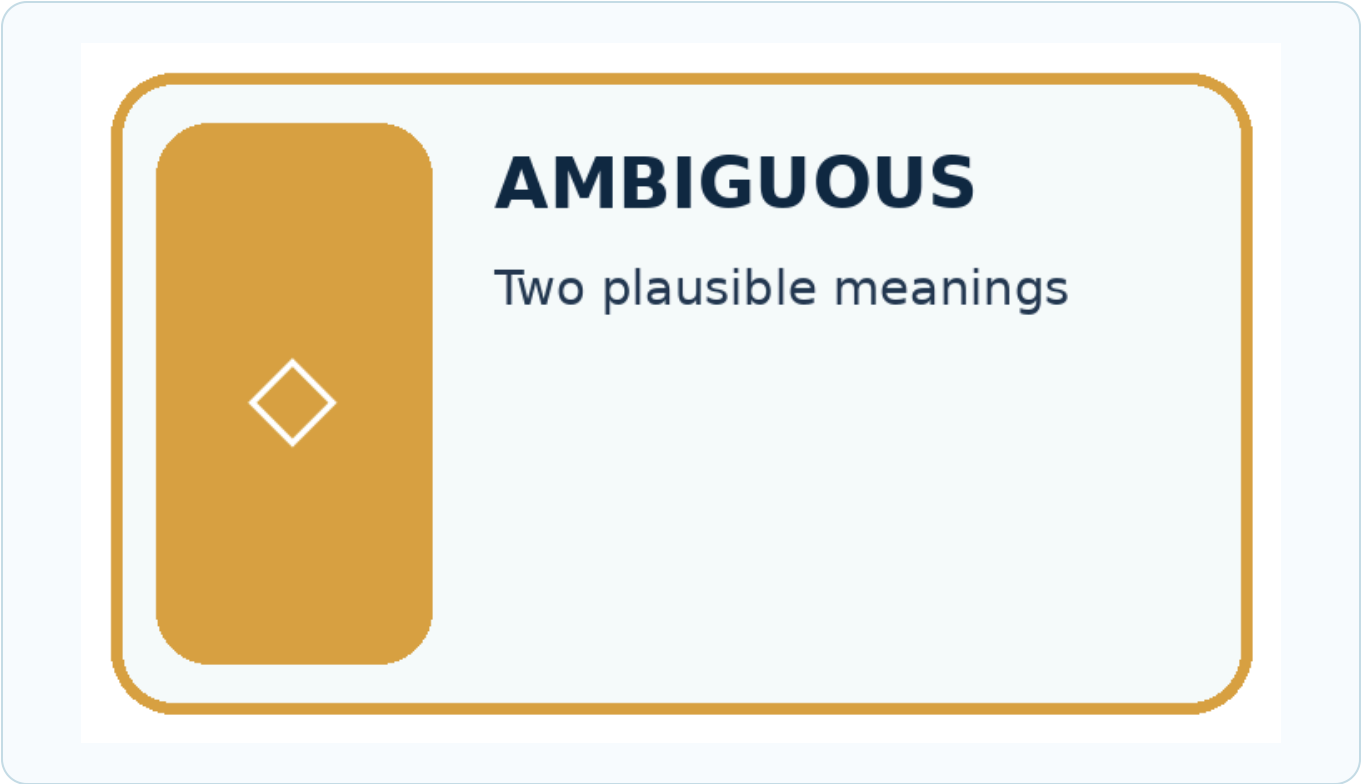
Compare case coverage with a flattering total

> Changing the case mixture can raise the total without fixing an existing failure.

Four pictures with a purpose



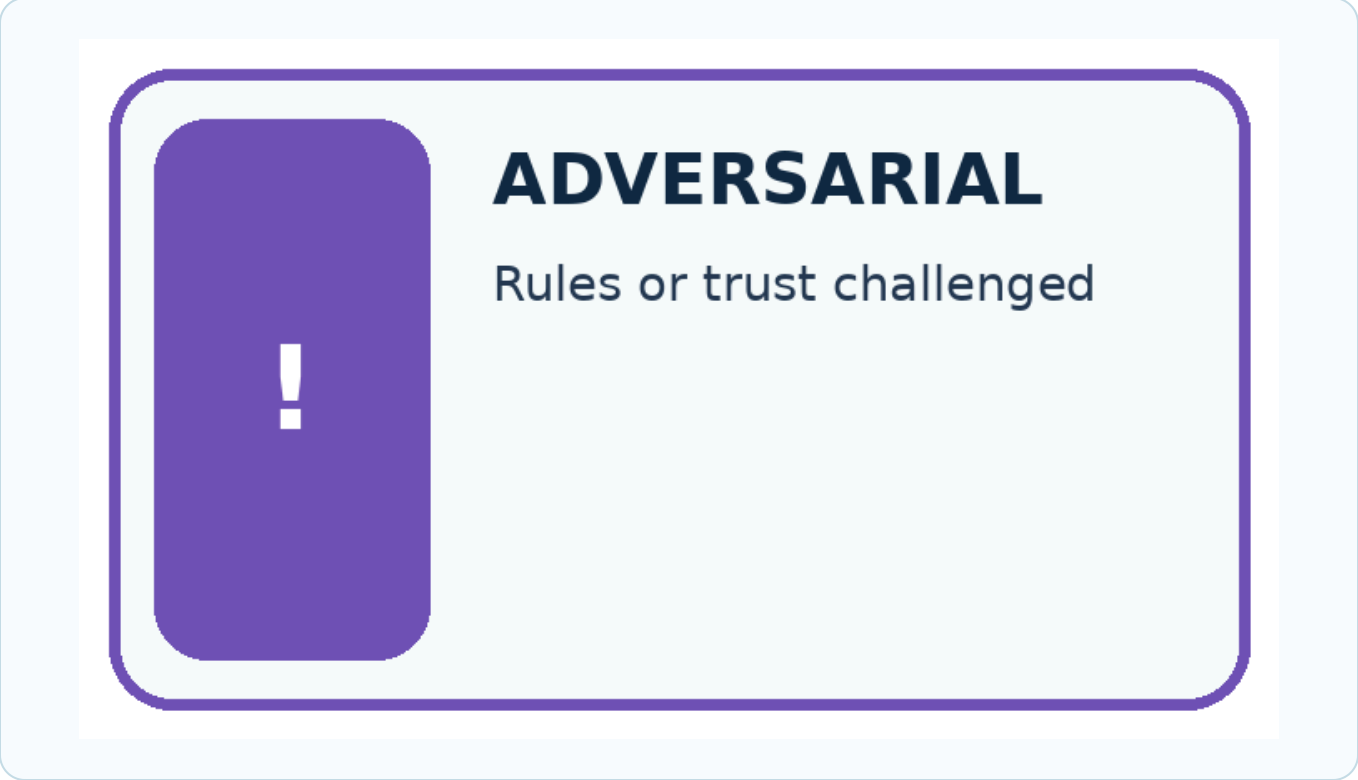
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Add three missing-**evidence** cases that all pass.

▼ Check after predicting

Prediction: Overall becomes 8/9, but supported answering remains 2/3.

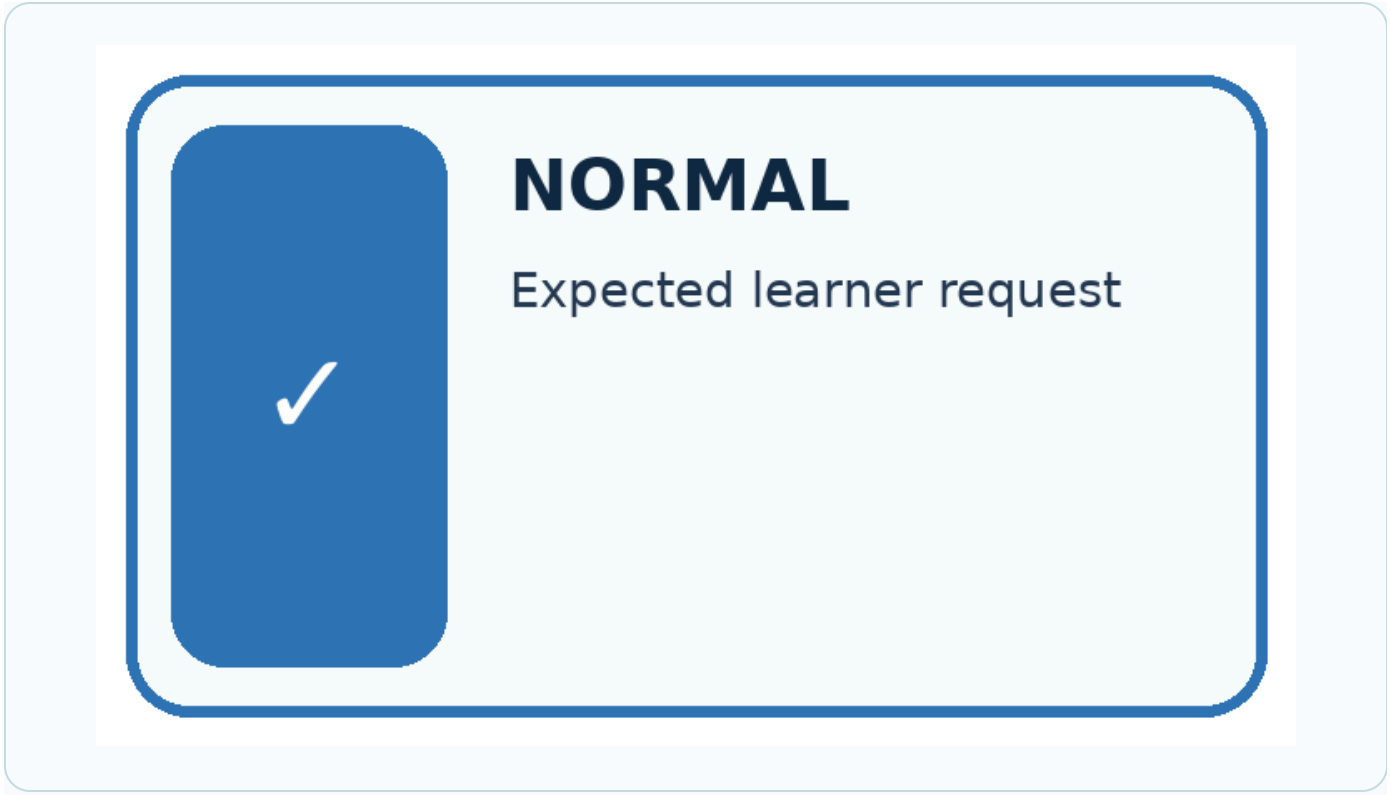
Reason: Changing the case mixture raises the pooled rate without improving any existing answerable result.

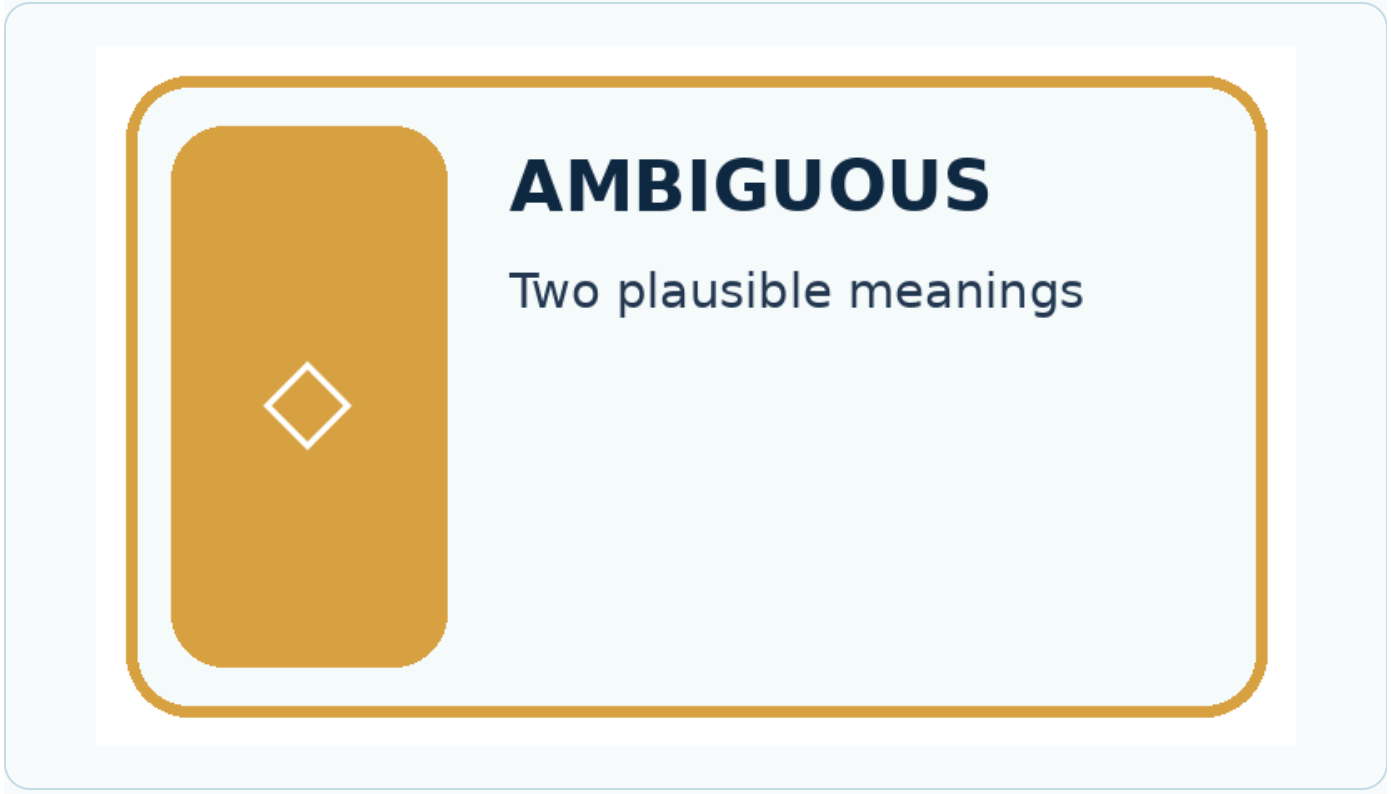
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace why the pooled rate changes when the cases change

> A better-looking total is not evidence of better answering.

Read the complete six-case outcome record

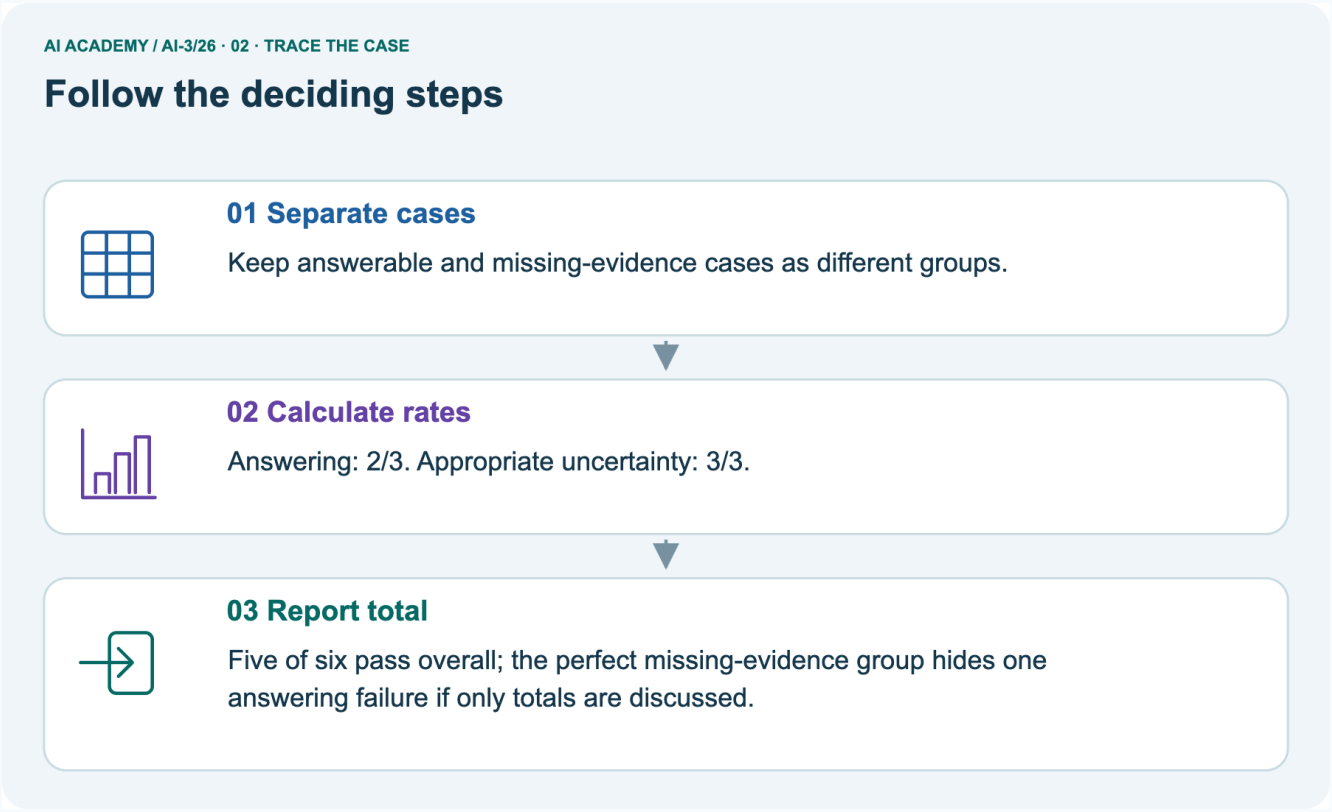
Design an **evaluation set** that checks answering and appropriate uncertainty separately.

Six toy questions: three have source answers and three do not. Results: 2 supported answers out of 3; 3 correct “not in source” responses out of 3.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Keep three records separate: the pictured six-case record has answering 2/3 and fallback 3/3, giving 5/6. Adding three successful missing-evidence cases gives 8/9 overall while answering remains 2/3. The separate ten-case W2 has answering 5/6 and fallback 3/4, giving 8/10. No system was run; these are supplied classroom outcomes, not estimated future accuracy.

Trace subgroup and pooled denominators



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Separate cases	Keep answerable and missing-evidence cases as different groups.
2. Calculate rates	Answering: 2/3. Appropriate uncertainty: 3/3.
3. Report total	Five of six pass overall; the perfect missing-evidence group hides one answering failure if only totals are discussed.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

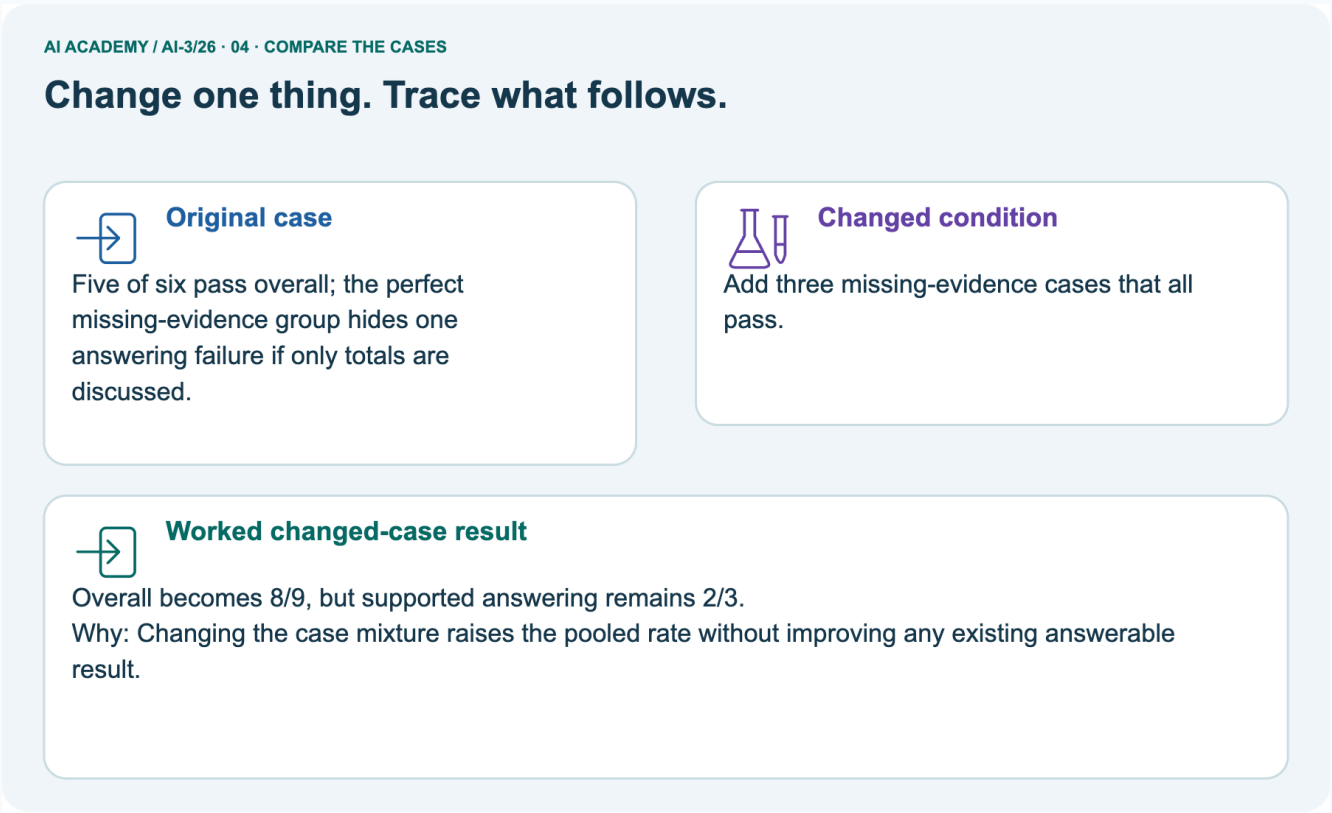
Read the picture in order. Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict adding only missing-evidence successes

- Change one thing: Add three missing-**evidence** cases that all pass. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State the constructed-set limit

- A small constructed set probes specific behaviours; its pooled rate is not an estimate of every future classroom question.

- Read the labelled picture: Did the added cases demonstrate better answering? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

▼ Separate practice · Ten-case W2

Ten cases include six answerable and four missing-**evidence** cases. System answers five answerable correctly and abstains appropriately on three missing cases.

Answer correctness 5/6; appropriate fallback 3/4; combined **contract pass**es 8/10 if each case has one required outcome. Different denominators answer different questions.

▼ Extension · Known relevance and top-three scores

Relevant documents are A, C, E. Top three results A, B, C contain two relevant items.

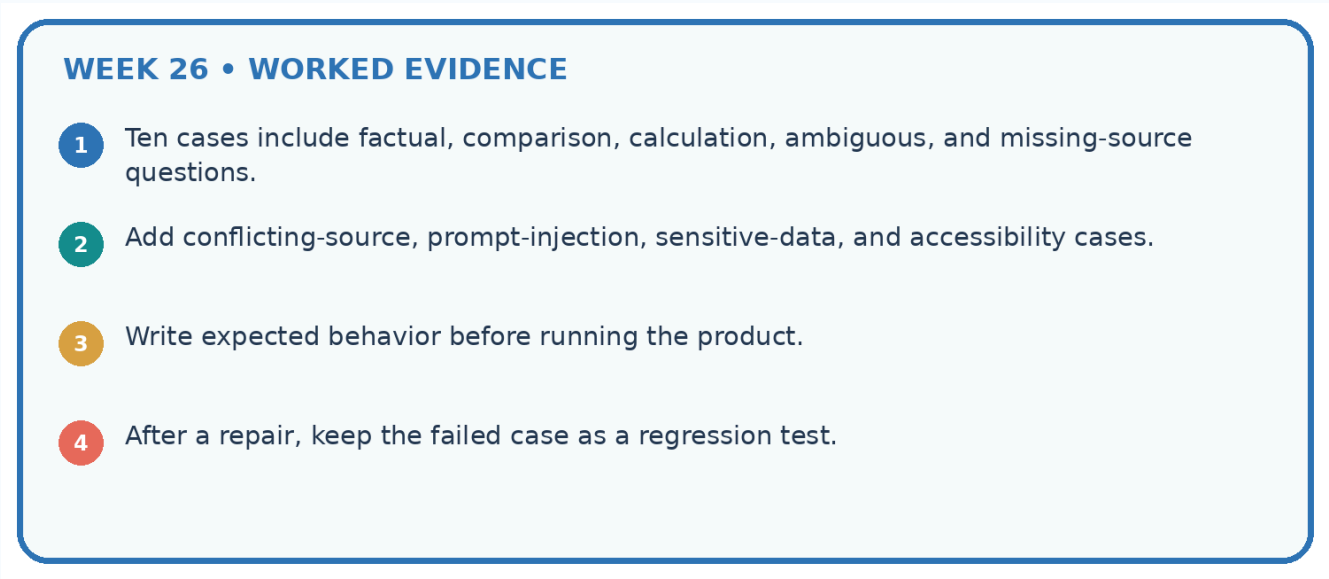
Precision@3=2/3; recall@3=2/3. Duplicates should not count as distinct retrieved documents.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Ten cases include factual, comparison, calculation, ambiguous, and missing-source questions.

2. Add conflicting-source, prompt-injection, sensitive-data, and accessibility cases.

3. Write **expected behavior** before running the product.

4. After a repair, keep the failed case as a **regression test**.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Apply the connected idea: Ranking and retrieval evaluation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Evaluation Sets

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find the changed denominator or contaminated test

> A pass count needs case identities, categories and exposure history.

Repair the pooled-improvement shortcut

Claim to repair: Five of six became eight of nine, so the assistant now answers source questions better. Only the case mixture changed: answering remains two of three. Preserve all original outcomes and show subgroup rates. Another shortcut is editing a prompt until it passes these cases, then calling the same reviewed cases a final test. Keep them as development evidence and use new reserved cases for a later check.

Design a revealing follow-up

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: More random test cases always create better coverage. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Apply the connected idea: Ranking and retrieval evaluation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

MAIN LESSON · SECTION 8

Practise the ten-case denominators with less help

> Do not merge this record with the pictured six-case record.

Complete the current W2

Use workbook W2 for the connected example “System **evaluation set**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Audit an all-future-cases claim

- Use the worked case for System **evaluation set**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the case-mixture change

- Did the added cases demonstrate better answering? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
> | |
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Keep independent work separate

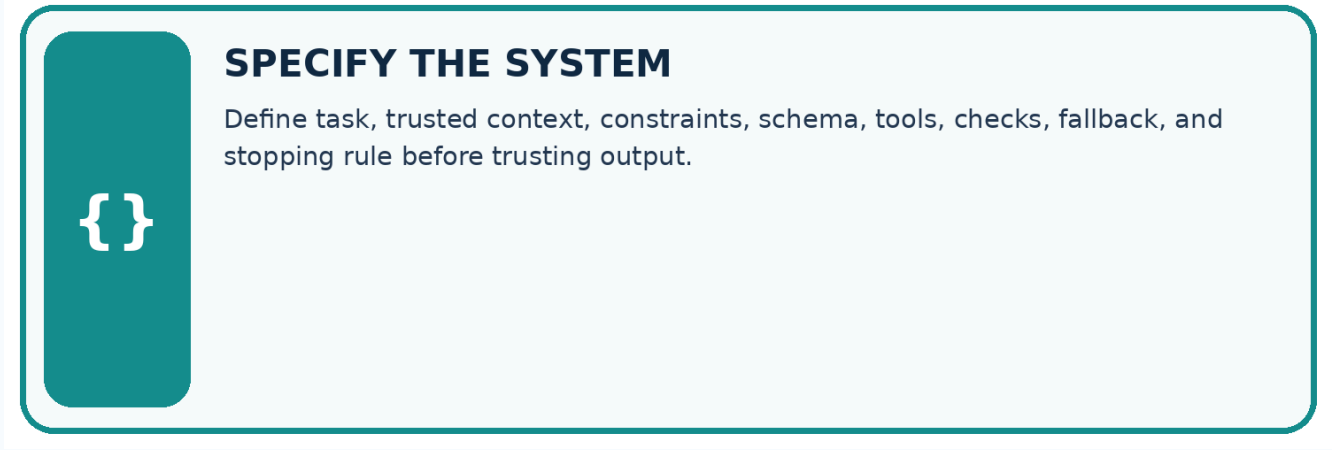
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

Evaluation **coverage** limits the claim; prompts tuned on these cases need new reserved final cases.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Apply the connected idea: Ranking and retrieval evaluation

Known judgments may be incomplete; a score over one query does not establish **coverage** over all queries.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Practise with less help: Trace the worked case

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Practise the connected topic: Ranking and retrieval evaluation

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: The worked result establishes every future case

Explain the picture

MAIN LESSON · SECTION 9

Audit ten complete cases, then find missing coverage

> The supplied outputs instantiate the W2 denominators.

All permitted source records

Eligible current public source	Complete fictional content
H-v1 · Library hours	Monday–Thursday opens 09:00, closes 17:00. Friday opens 09:00, closes 16:00.
R-v1 · Return instructions	Return books in the blue box.
C-v1 · Room notice	Room B is approved for this activity; capacity 12.
Missing facts	No manager name, lunch provision, bus time or weekend opening is supplied.

Ten questions, expected outcomes and constructed outputs

ID / question	Expected behavior	Supplied constructed output
A1 · Monday opening	09:00 from H-v1	09:00 [H-v1]
A2 · Tuesday closing	17:00 from H-v1	17:00 [H-v1]
A3 · Friday closing	16:00 from Friday clause	16:00 [H-v1]
A4 · Return location	Blue box from R-v1	Red box [R-v1]

ID / question	Expected behavior	Supplied constructed output
A5 · B capacity	12 from C-v1	12 [C-v1]
A6 · Is B approved?	Yes under C-v1 scope	Approved for this activity [C-v1]
U1 · Manager name	Not supplied; request source	Not in these sources.
U2 · Lunch provision	Not supplied; request source	Free lunch is included.
U3 · Bus time	Not supplied; request source	Not in these sources.
U4 · Saturday opening	Not supplied; request source	Not in these sources.

Score, inspect coverage and author new cases

1. Use the ten complete visible practice cases and frozen source versions. Treat A1–A6 as answerable and U1–U4 as missing-evidence cases. Judge each supplied output against its expected behavior, retaining A4 and U2 as failures. Compute 5/6 answer correctness, 3/4 appropriate fallback and 8/10 contract passes; each case has one required outcome in this exercise. These are constructed outputs, not a model run.

   Peer-audit coverage: this set still lacks ambiguity, conflicting authority and access cases. Author one fully supplied new case per missing category and freeze expectations, but do not invent observed results. Keep independent club-record cases withheld. Visible reviewed cases cannot become reserved final cases by relabelling them.

Keep the evaluation record

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the complete ten-case H/R/C practice set and supplied constructed outputs. Freeze source versions and expectations, score case-level outcomes, report subgroup rates, then identify missing ambiguity/conflict/access **coverage**. Newly authored cases have predictions, not fabricated results.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: design checks for fresh club-record gaps

> Distinguish ambiguity, absent facts and unresolved conflicting versions.

Independent notes, questions and scope

Approved notes: Robotics Tuesday Room2; Art Wednesday Room5. Requests: “Where does the club meet?” and “Who leads drama?” An unversioned added note says Robotics Room3. Design checks and safe responses.

Use only the supplied club notes and questions. The added unversioned note has no supplied precedence or authority rule. Do not assume a later-looking note overrides another, or invent a club leader. Design checks and expected responses; no actual system outcomes are supplied.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Changed independent case

Week 26 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

You improve a prompt after reading every test question. Can those questions still provide an untouched final evaluation? Propose a repair.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Maintain a balanced eval set covering utility, grounding, format, privacy, and safety.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A test set contains ten easy questions with direct answers in the notes. Add three different kinds of cases that probe fallback behaviour and define expected outcomes.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project's evaluation set inspectable

> A case's expected answer and exposure history are part of the evidence.

This week's contribution

For the Verified Study Buddy, maintain a versioned case table with source eligibility, expected behavior, criterion decisions, actual output and exposure role. Keep subgroup failures visible even when the pooled total is high. Separate debugging cases from new reserved evaluation. If tuning after a failed final case, record the exposure and use further fresh cases before making a new generalisation claim.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Retain the checkpoint and human choice

THIS WEEK'S CHECKPOINT Maintain a balanced eval set covering utility, grounding, format, privacy, and safety.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

> **Write or draw your reasoning and evidence.**

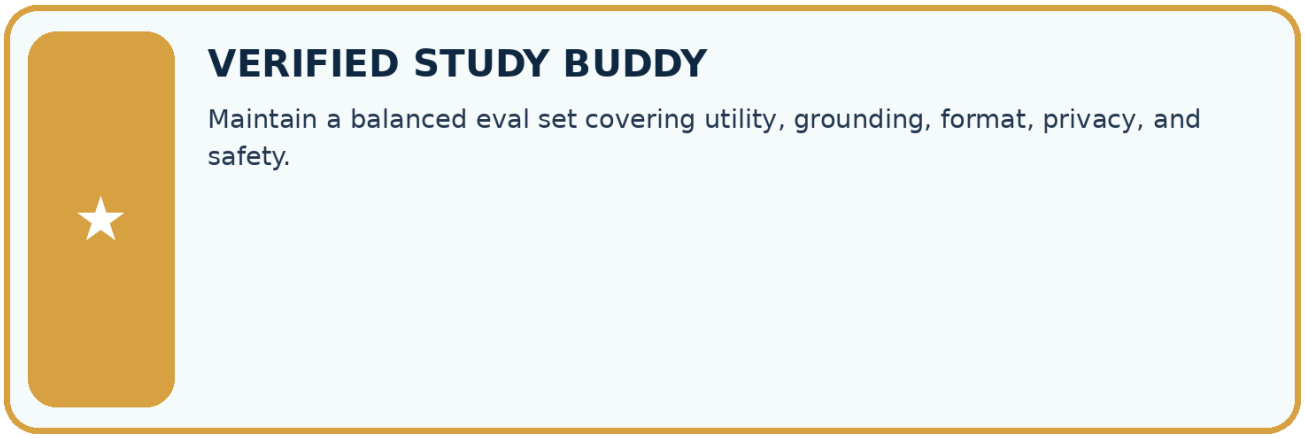
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: name the subgroup, denominator and untested risk

> A visible practice set cannot prove every future case.

Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of system **evaluation set**s

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the main lesson and extension**

REFERENCE · SECTION 5

Reference · Words for the main lesson and extension

> Use each word with a concrete record or calculation.

Core words

Word	My own example	Not this / boundary
evaluation set	Cases used to measure a system under declared criteria; specify whether they are validation or final test cases.	
coverage	The range of relevant cases, conditions or groups included in data or tests.	
expected behavior	The result or action required by the specification for a given case, stated before observing the system.	
regression test	A repeatable check that previously correct behaviour still works after a change.	

Main explanation terms

Word	Meaning
Evaluation case	An input with evidence and prespecified expected behaviour

Word	Meaning
Contract pass	Satisfying the declared requirements for one case

Extension terms

Term	Meaning
Precision at k	Relevant results among the first k returned items
Recall at k	Fraction of known relevant eligible items found in the first k results
Cutoff	The rank limit used for scoring

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
COVERAGE
important situations included
3
EXPECTED BEHAVIO
observable acceptable response



Words used in the closer explanation

Terms for Ranking and retrieval evaluation

▼ **Extra challenge · Choose a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Choose a stronger explanation

> Choose the next useful challenge rather than repeating the ladder.

Thinking ladder

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.

- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest **evidence** of independent understanding:

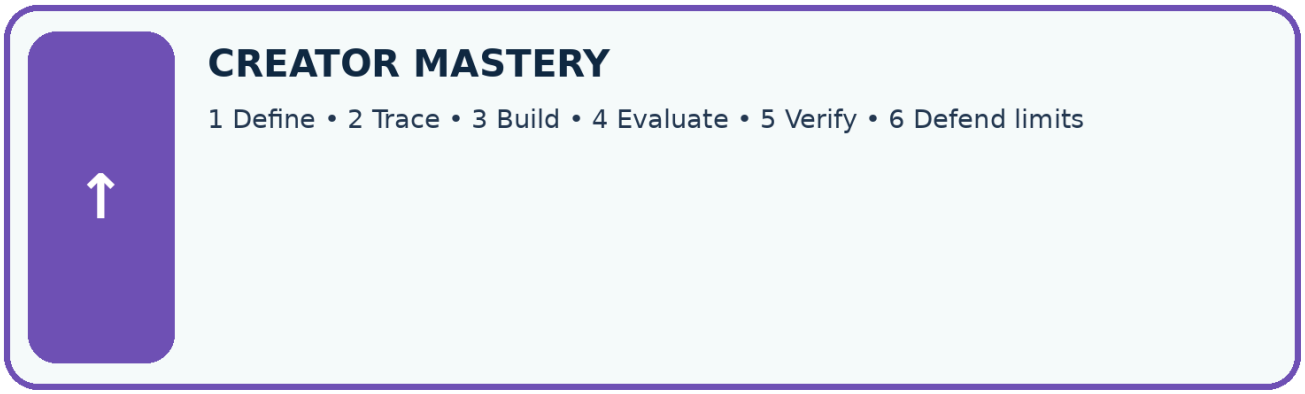
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference · Check your explanation

Reference · Check your explanation

> Use after practice; Design checks for the fresh club records independently is the selected independent task.

Evidence and limits

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Self-check

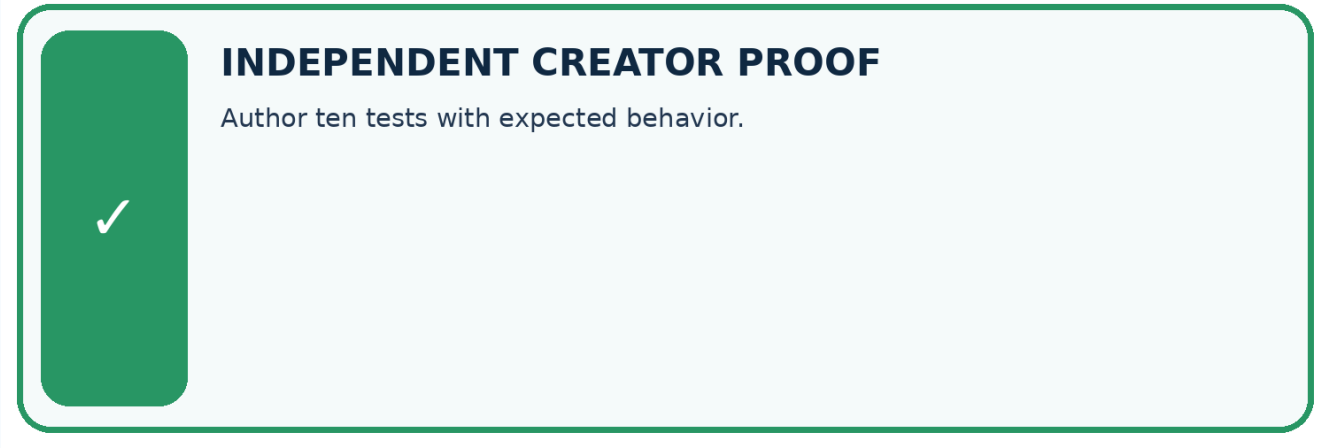
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer ‘why?’

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT MASTERY PROOF

Checklist for this week’s explanation

Use after practice. Design checks for the fresh club records independently is the selected independent task.



Require clarification, missing-source behavior and conflict preservation before aggregation.

▼ Extra practice · The earlier complete case

REFERENCE · SECTION 13

Extra practice · The earlier complete case

> Try the question before opening the worked explanation.

Question and given facts

Ten evaluation questions are easy definitions; none lacks a source.

What is untested?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Absent **evidence**, ambiguity and boundary handling. Add predefined fallback cases and keep final cases separate from tuning.

Evidence or stage	Value or result
Easy definitions	10
Absent evidence	0
Ambiguity	0
Boundary handling	0

▼ Earlier wording, alternatives and picture transcripts

Week 26 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Empty evaluation categories reveal gaps

Ten evaluation questions are easy definitions; none lacks a source.

What is untested?

Absent **evidence**, ambiguity and boundary handling. Add predefined fallback cases and keep final cases separate from tuning.

▼ **Reference · Earlier checklist**

REFERENCE · SECTION 15

Reference · Earlier checklist

> Use when it helps inspect the evidence.

Checklist and first prediction

By the end, I can	How I will show it
Explain the mechanism and terminology of system evaluation sets	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: Design an **evaluation set** that checks answering and appropriate uncertainty separately.

▼ **Extra practice · Pictures, feedback and repair**

REFERENCE · SECTION 17

Extra practice · Pictures, feedback and repair

> Try before opening feedback; keep the selected independent task separate.

First supported question

- Design an **evaluation set** that checks answering and appropriate uncertainty separately.
- Which labelled step matches this detail? Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed.
- 1. Separate cases
- 2. Calculate rates
- 3. Report total

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

- After your attempt, compare with the supported explanation: Report total Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed. Read the picture in order.

  Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after trying**

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: Add three missing-**evidence** cases that all pass. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. Overall becomes 8/9, but supported answering remains 2/3.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

- After your attempt, compare with the supported explanation: Overall becomes 8/9, but supported answering remains 2/3. Changing the case mixture raises the pooled rate without improving any existing answerable result.

> **My selected conclusion, deciding fact and explanation**

▼ **Check changed case**

Repair a missing building block

Repair and keep evidence

Review Quality Rubrics (ai-3/25). Goal: Judge an output using separate observable rubric criteria. Short practice: Why is “one out of two checks passed” an insufficient release decision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Block to Python Bridge (ai-3/23). Goal: Translate a short paper calculation into a Python script. Short practice: Trace the changed list and identify where the result differs.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Conversation State and Memory (ai-3/20). Goal: Explain the **mechanism** and terminology of conversation state and retention. Short practice: What is selected_book after the correction and after exit?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

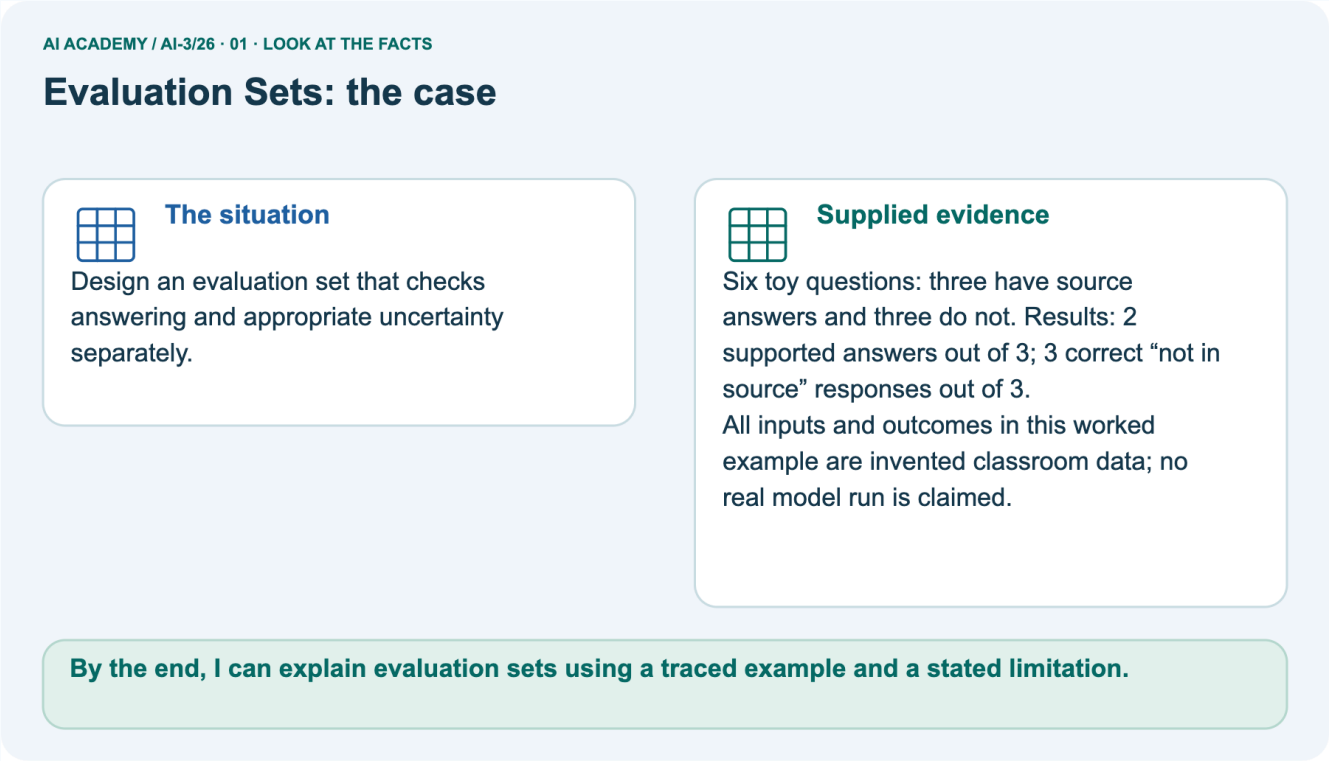
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Revisit the picture that serves the current question.

01. Evaluation Sets: the case



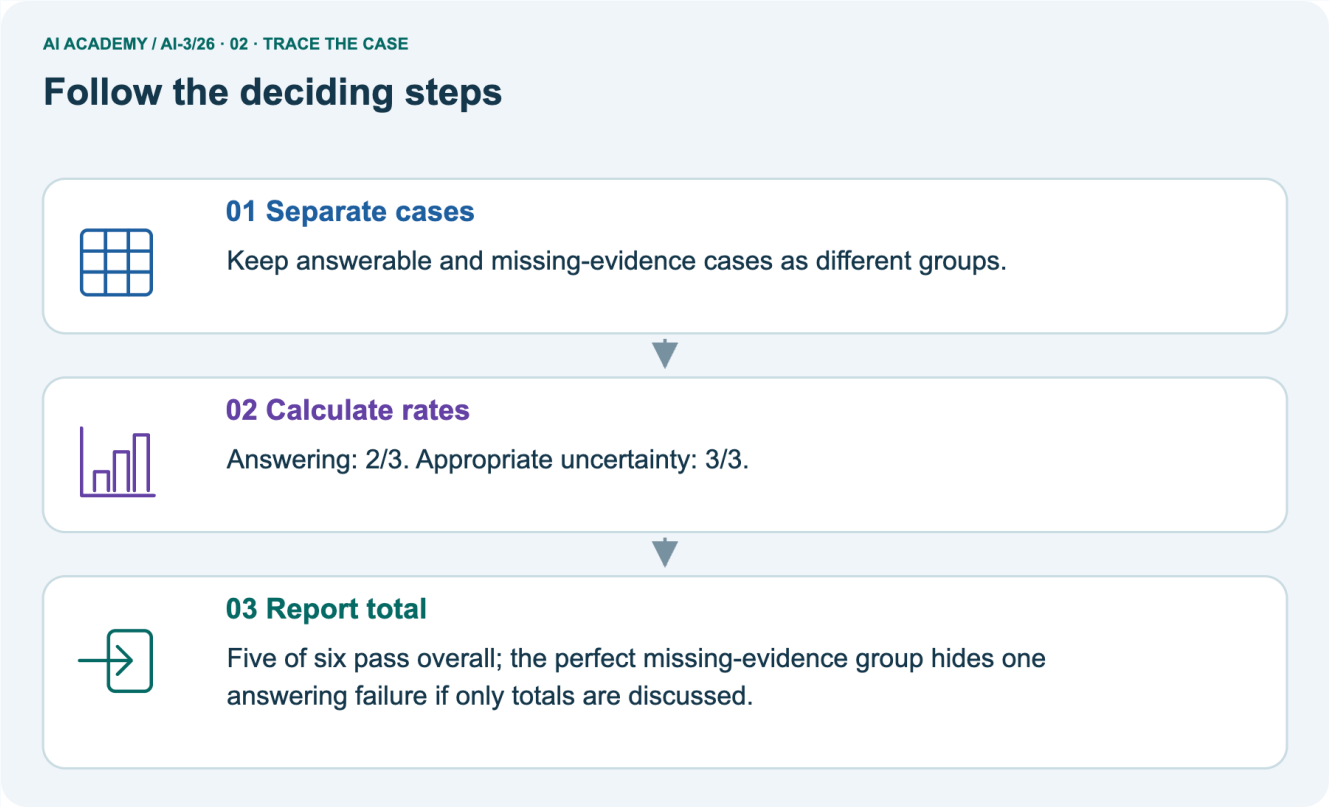
Follow the labelled picture with the matching explanation. Open the vector diagram

Design an **evaluation set** that checks answering and appropriate uncertainty separately. Supplied facts: Six toy questions: three have source answers and three do not.

Results: 2 supported answers out of 3; 3 correct “not in source” responses out of 3. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



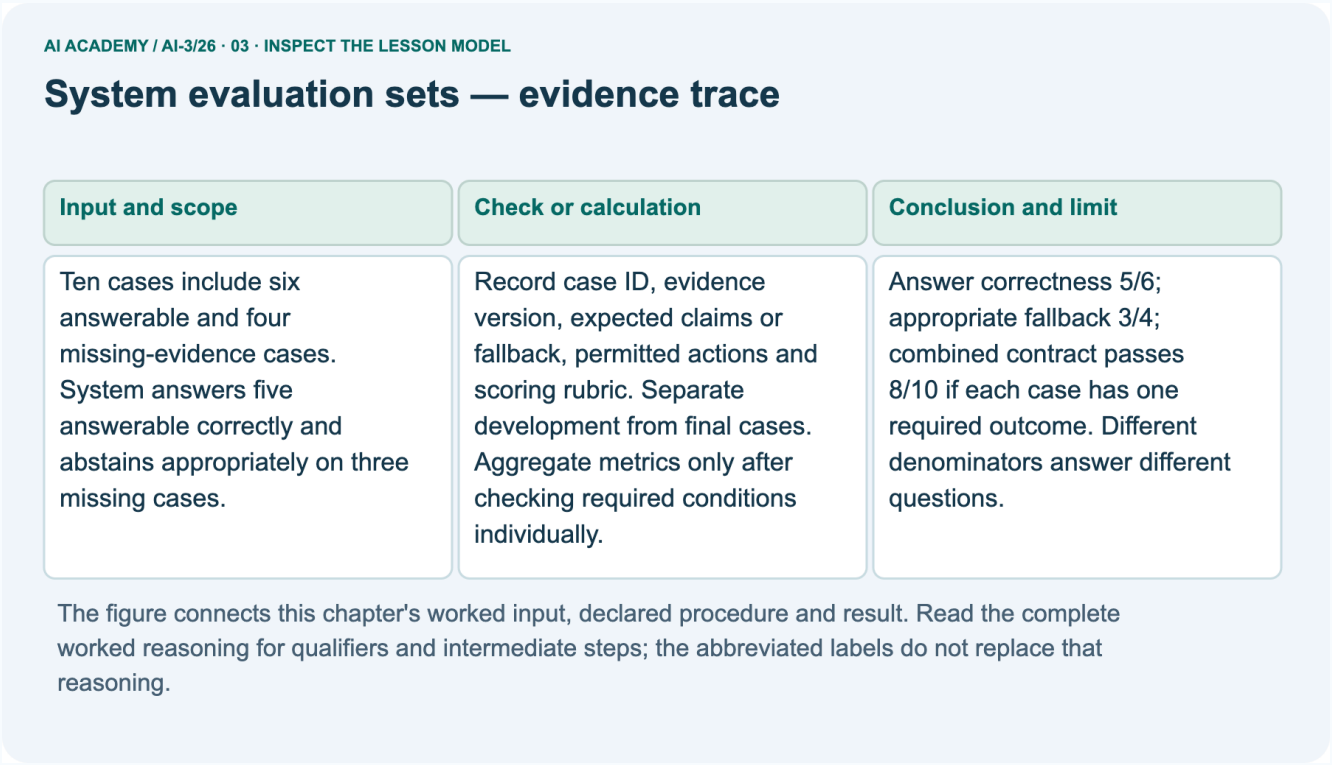
Follow the labelled picture with the matching explanation. Open the vector diagram

Separate cases: Keep answerable and missing-**evidence** cases as different groups. → Calculate rates: Answering: 2/3.

Appropriate uncertainty: 3/3. → Report total: Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed.

Point to each numbered stage and explain which recorded **input** it uses.

03. System evaluation sets — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

System **evaluation set**s — **evidence** trace. Ten cases include six answerable and four missing-**evidence** cases.

System answers five answerable correctly and abstains appropriately on three missing cases. | Record case ID, **evidence** version, expected claims or fallback, permitted actions and scoring rubric.

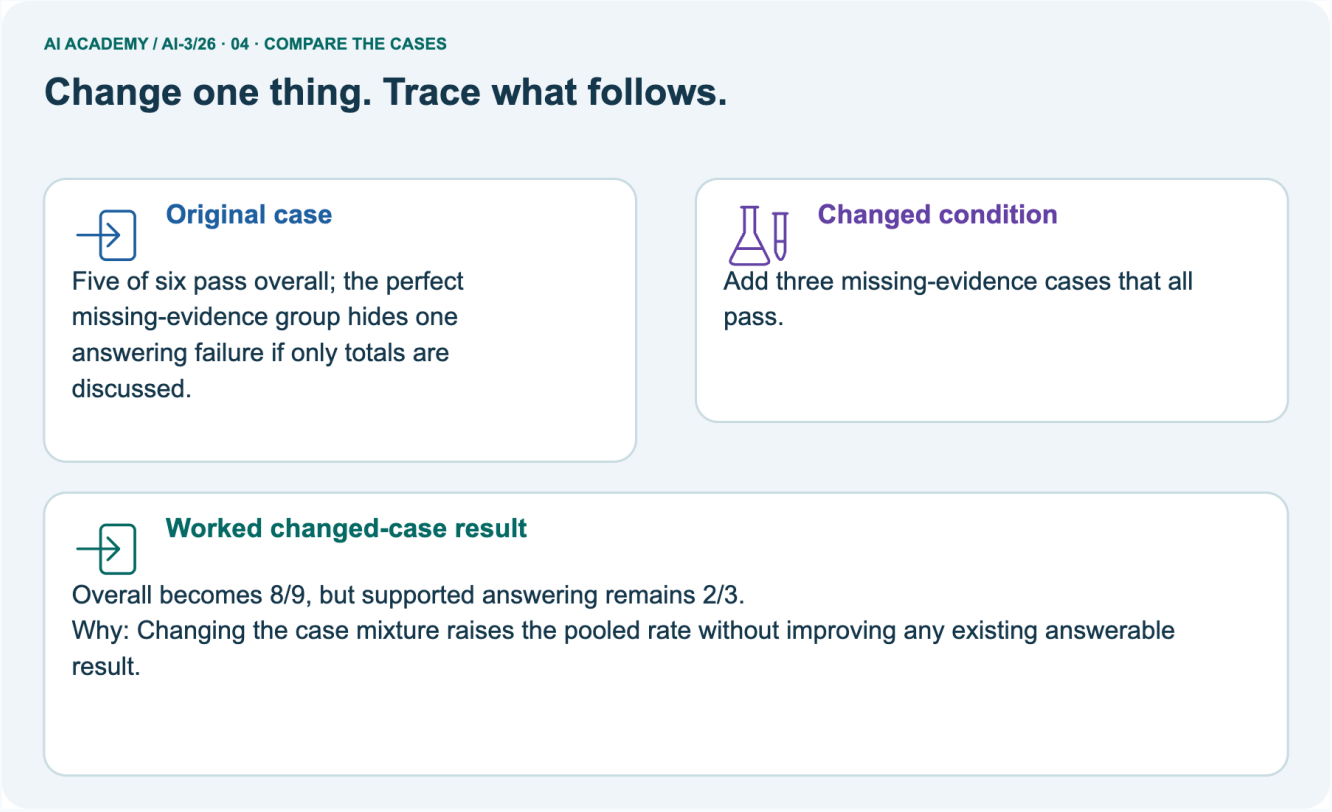
Separate development from final cases.

Aggregate metrics only after checking required conditions individually. | Answer correctness 5/6; appropriate fallback 3/4; combined **contract pass**es 8/10 if each case has one required outcome.

Different denominators answer different questions.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



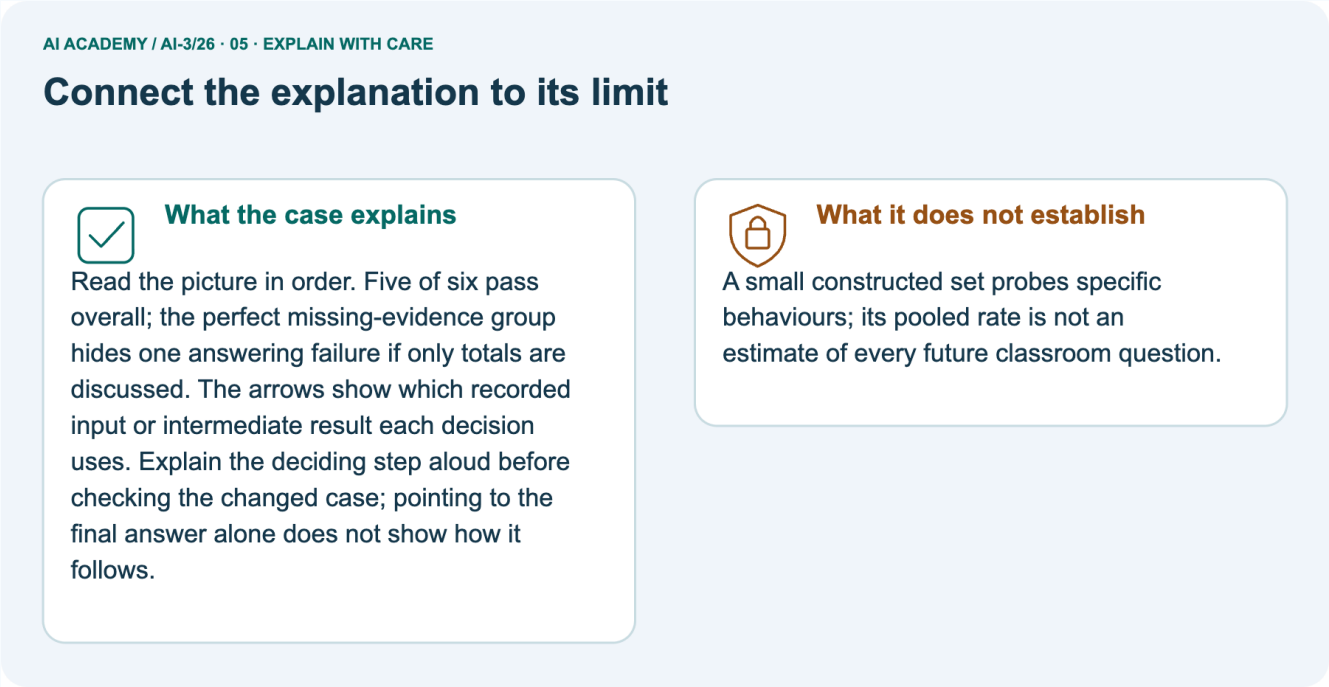
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed. Changed condition: Add three missing-**evidence** cases that all pass.

Expected result: Overall becomes 8/9, but supported answering remains 2/3. Reason: Changing the case mixture raises the pooled rate without improving any existing answerable result.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

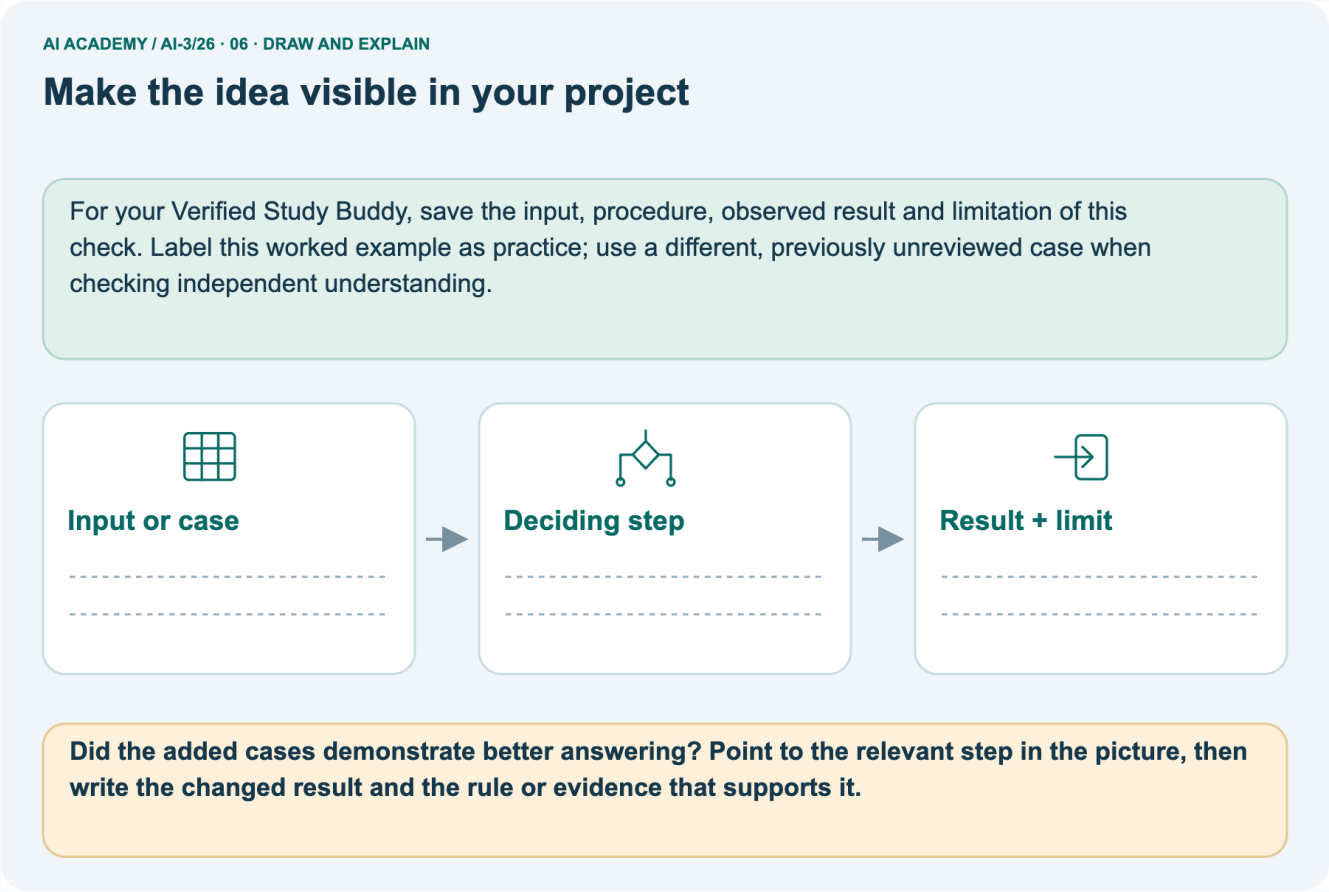
Read the picture in order. Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A small constructed set probes specific behaviours; its pooled rate is not an estimate of every future classroom question.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Did the added cases demonstrate better answering? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Evaluation Sets: the case

02. Follow the deciding steps

03. System evaluation sets — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.